

Effect of tomato industrial processing on phenolic profile and antiplatelet activity.

BACKGROUND: Regular consumption of fruits and vegetables (e.g., tomatoes) has been shown to be beneficial in terms of reducing the incidence of cardiovascular diseases. The industrial processing of tomatoes into tomato-based products includes several thermal treatments. Very little is known on the effect of tomato industrial processing on antiaggregatory activity and phenolic profile. **METHODS:** It was assessed the effect of tomato and by-products extracts on platelet aggregation induced by ADP, collagen, TRAP-6 and arachidonic acid. These in vitro antithrombotic properties were further supported in an in vivo model of thrombosis. A set of antiplatelet compounds has been selected for HPLC analysis in the different extracts. **RESULTS:** Some natural compounds such as chlorogenic, caffeic, ferulic and p-coumaric acids were identified by HPLC in tomatoes and its products may inhibit platelet activation. Red tomatoes, tomato products (sauce, ketchup and juice) and by-products extracts inhibited platelet aggregation induced adenosine 5'-diphosphate, collagen, thrombin receptor activator peptide-6 and arachidonic acid, but to a different extent. Also, pomace extract presents antithrombotic activity. **CONCLUSIONS:** Processed tomatoes may have a higher content of health-benefiting compounds than fresh ones. Pomace even presents the best antiplatelet activity. Finally, tomato products may be used as a functional ingredient adding antiplatelet activities to processed foods.